

L27

Sep 28, 2026

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Messages from the Last Class

- **Message 1: Bias and precision**

Accuracy/Bias: how close/far, on an average, the dart throws are to the bullseye

Precision: how close dart throws are to each other

$$\text{bias} = E[\hat{\theta} - \theta]$$

- **Message 2: Parameter estimation (MoM)**

Observation: calculate the ^{observed} moments $\Rightarrow$ eg, mean & variance
Equate them to the distribution's moments.
Solve the system of equations to obtain
Model parameters.

Messages from the Last Class

• Message 3: Parameter estimation (MLE)

Maximum Likelihood Estimation

$$\mathcal{L}(\mu, \sigma | x_i) = f_x(x_i | \mu, \sigma)$$

θ : parameters

$$(\hat{\mu}, \hat{\sigma}) = \arg \max \ln \mathcal{L}(\mu, \sigma | X)$$

$$\frac{\partial \ln \mathcal{L}}{\partial \mu} = 0$$

$$\frac{\partial \ln \mathcal{L}}{\partial \sigma} = 0$$

System of simultaneous

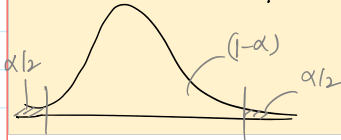
eqns.

Solve them $\hat{\mu}, \hat{\sigma}$

• Message 4: Confidence Interval (CI)

A $100(1-\alpha)\%$ confidence interval is

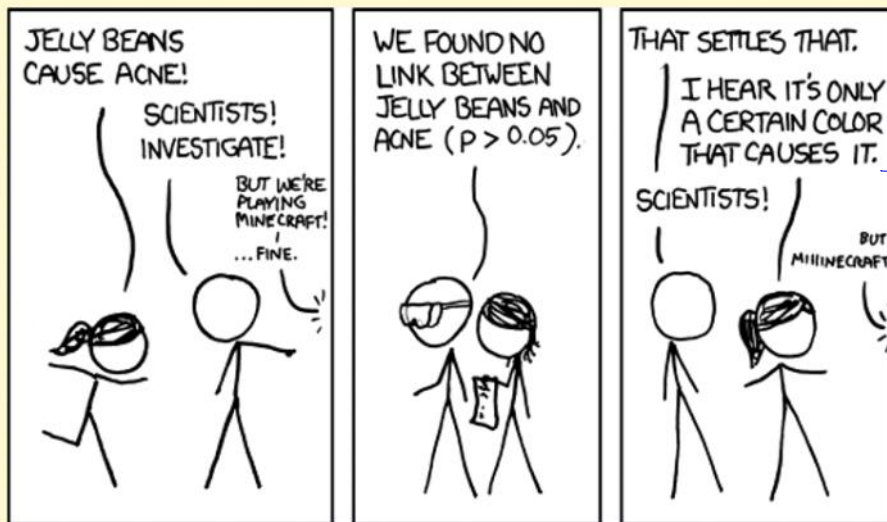
$$\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$



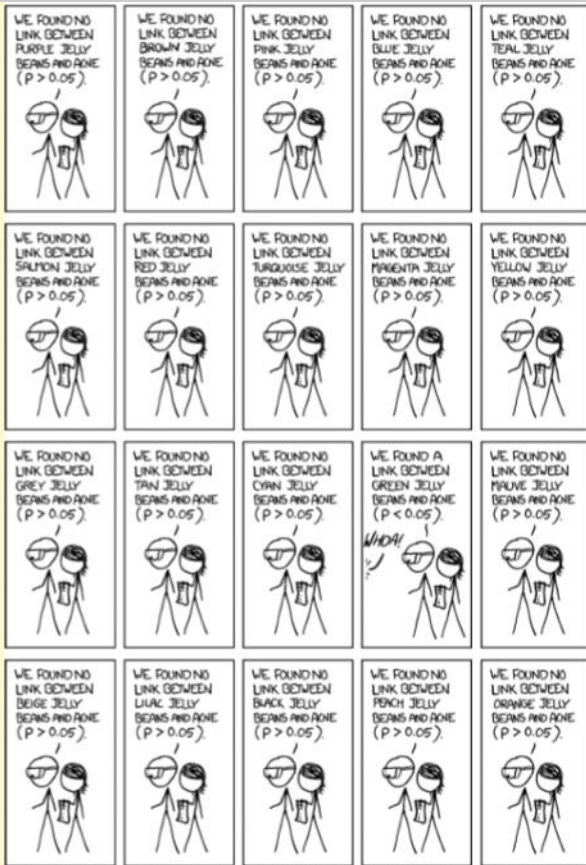
xkcd

a webcomic of Romance, sarcasm, math, & language.

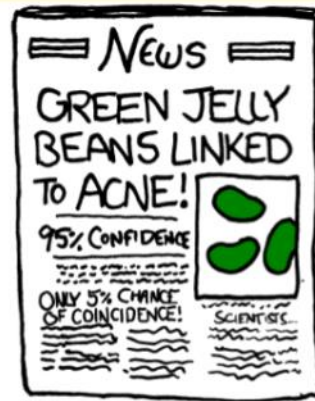
WHAT IF-2.



<https://xkcd.com/882>



<https://xkcd.com/882>
<https://explainxkcd.com/882>

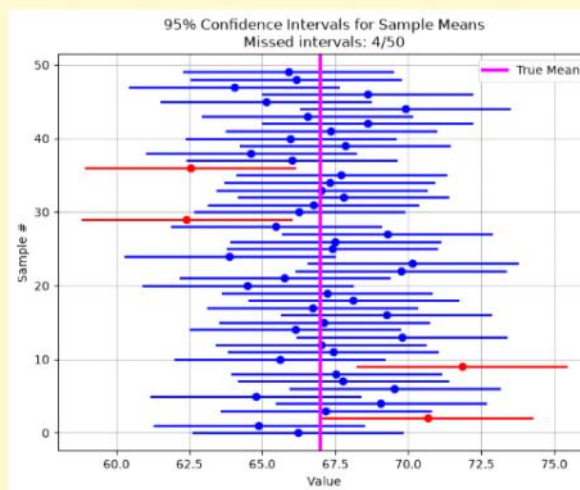


Messages from the Last Class

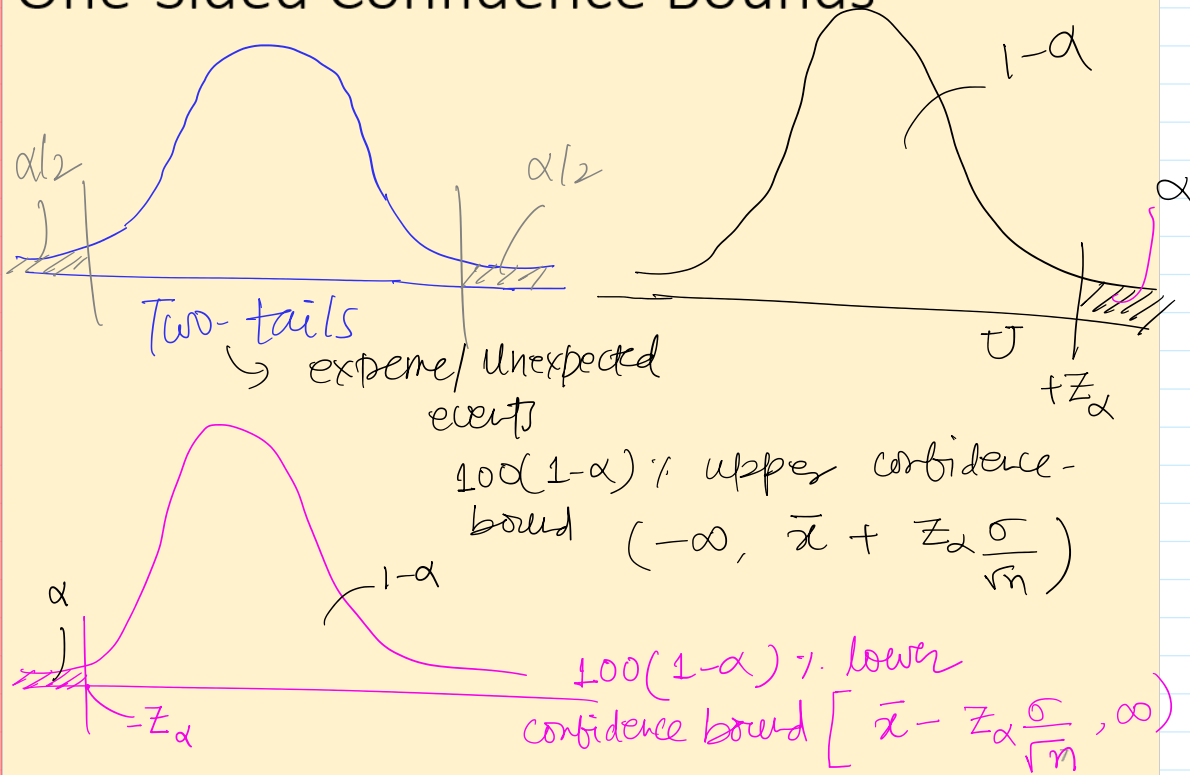
• Message 5: Parameter estimation (MLE)

Probability that μ is in $[\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}]$ is 95%.

Tempting but incorrect. as μ is NOT a σ .



One-Sided Confidence Bounds



Small Sample Sizes

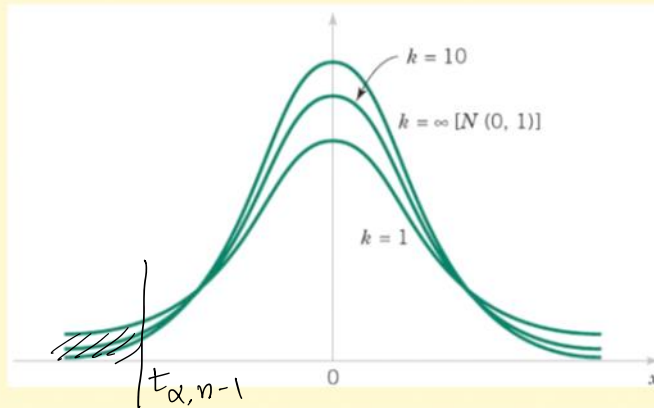
Small sample sizes

When sample size, n is small.

$\bar{x}$ follows t -dist. with $(n-1)$ dof.

t -dist. approaches $\mathcal{N}$ when $n \geq 30$.

$$\Pr \left[\bar{x} - t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} \right] = 1 - \alpha$$



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t Table

t Table		t _{.50}	t _{.75}	t _{.80}	t _{.85}	t _{.90}	t _{.95}	t _{.975}	t _{.99}	t _{.995}	t _{.999}	t _{.9995}
cum. prob		0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
one-tail		1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
two-tails		1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df												
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62	
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599	
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924	
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610	
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869	
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959	
7	0.000	0.711	0.896	1.119	1.415	1.885	2.365	2.998	3.490	4.785	5.408	
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041	
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781	
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.784	3.169	4.144	4.587	
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437	
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318	
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221	
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140	
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073	
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.688	4.015	
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965	
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922	
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883	
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850	
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819	
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792	
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768	
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745	
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725	
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707	
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690	
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674	
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659	
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646	
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551	
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.680	3.232	3.460	
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.418	
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390	
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300	
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291	
		0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
		Confidence Level										

Normal →

Unknown Variance

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Known and Unknown Variance (CI for μ)

Pop Var. σ^2	Sample Size n	Test Statistic & Sampling Dist.	Std. Error SE	C.I.	Assumption
Known (σ given)	Small $n < 30$	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	$\frac{\sigma}{\sqrt{n}}$	$[\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}]$	population must be NORMAL
Known (σ given)	Large $n \geq 30$	$\downarrow$	$\downarrow$	$\downarrow$	population can be Non-Normal (Z valid by CLT)
Unknown (σ estimated by s)	Small $n < 30$	$T = \frac{\bar{X} - \mu}{s/\sqrt{n}}$ $\sim T_{n-1}$		$[\bar{x} - t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}, \bar{x} + t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}]$	
Unknown (σ estimated by s)	Large $n \geq 30$	$Z = \frac{\bar{X} - \mu}{s/\sqrt{n}}$ $\sim N(0, 1)$		$[\bar{x} - z_{\alpha/2} \frac{s}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{s}{\sqrt{n}}]$	Z valid by CLT $s \rightarrow \sigma$ valid by LLN Population Variance
C.L.T. $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$ as $n \rightarrow \infty$		Law of Large Numbers (LLN) $s \xrightarrow{p} \sigma$ as $n \rightarrow \infty$			

standard error

Confidence interval for σ^2

Confidence interval for σ^2

- Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with mean μ and stdev σ . Let s^2 be the sample variance.

$\frac{(n-1)s^2}{\sigma^2}$ follows χ^2 distribution with $(n-1)$ dof

$$\chi^2_{1-\frac{\alpha}{2}, n-1} \leq \frac{(n-1)s^2}{\sigma^2} \leq \chi^2_{\frac{\alpha}{2}, n-1}$$

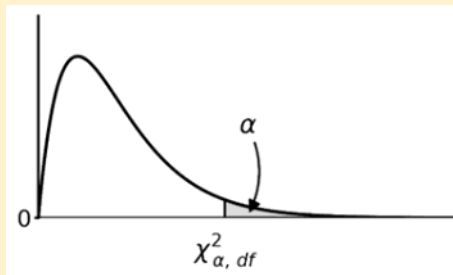
$$\Rightarrow \frac{(n-1)s^2}{\chi^2_{\frac{\alpha}{2}, n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\frac{\alpha}{2}, n-1}}$$

$$\bar{X} \sim \mathcal{N} \quad \frac{(\bar{X} - \mu)^2}{\frac{(n-1)s^2}{\sigma^2}} \sim \chi^2_{n-1}$$

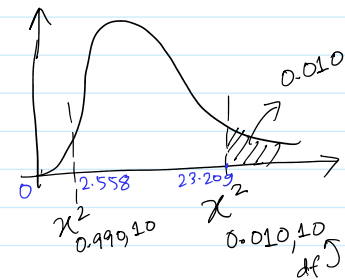
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Example-3 on CI for σ^2

- $n = 10$ measurements of concrete cylinder compressive strength (in MPa) are recorded as follows:
32.1, 33.4, 31.8, 32.5, 33.0, 32.7, 31.9, 33.2, 32.4, 32.0
- The sample variance is calculated as $s^2 = 0.33 \text{ MPa}^2$
- Assuming compressive strength is normally distributed, find a **95% confidence interval** for:
- The population variance σ^2
- The population standard deviation σ



df	$\chi^2_{.995}$	$\chi^2_{.990}$	$\chi^2_{.975}$	$\chi^2_{.950}$	$\chi^2_{.900}$	$\chi^2_{.800}$	$\chi^2_{.700}$	$\chi^2_{.600}$	$\chi^2_{.500}$	$\chi^2_{.400}$	$\chi^2_{.300}$	$\chi^2_{.200}$	$\chi^2_{.100}$	$\chi^2_{.050}$	$\chi^2_{.025}$	$\chi^2_{.010}$	$\chi^2_{.005}$
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879							
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597							
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838							
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860							
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750							
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548							
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278							
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955							
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589							
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188							
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757							
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300							
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819							
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319							
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801							
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267							
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718							
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156							
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582							
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997							
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401							
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796							
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181							
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559							
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928							
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290							
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645							
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993							
29	13.121	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336							
30	13.787	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672							
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766							
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490							
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952							
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215							
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321							
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299							
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169							



M2C2

Interval Estimation: two-sided and one-sided CI for population μ and σ^2 under known and unknown variance assumptions.

Hypothesis Testing Principles: H_0 and H_1 .

Error Analysis & Test Power: Type I (α) and Type II (β) error probabilities, p -values.

Parametric Testing: Z -tests, t -tests, χ^2 -tests, and F -tests for single-sample and two-sample population parameters.

Empirical Probability Plotting: probability paper and ECDFs to evaluate candidate dist. visually.

Goodness-of-Fit Verification: χ^2 , Kolmogorov-Smirnov (K-S), and Anderson-Darling (A-D) non-parametric hypothesis tests to validate distribution fits.

Motivation

How to test hypotheses and validate probability models?

M2C3

Ordinary Least Squares (OLS) Linear Regression: sum of squared errors, $SSE = \sum e_i^2$

Correlation & Model Evaluation: sample corr. Coeff. and R^2 to assess GoF.

Heteroscedasticity & Transformations: Weighted Least Squares (WLS) for non-constant Var. and parameter-linearizing transf. (e.g., Box-Cox) for nonlinear trends.

Multiple Linear Regression: Extend OLS to k -variable linear systems using matrix formulations ($\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$)

Surrogate Modeling & Probabilistic Machine Learning: Implement Gaussian Process Regression (Kriging) for rapid probabilistic surrogate modeling with spatial error bounds under uncertainty.

Motivation

How to map empirical relationships between variables to build predictive equations?

Thank you!